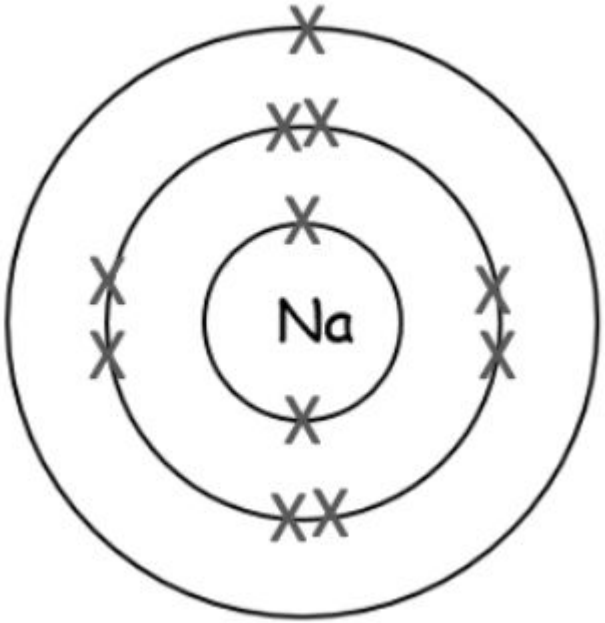
Starter - to discuss



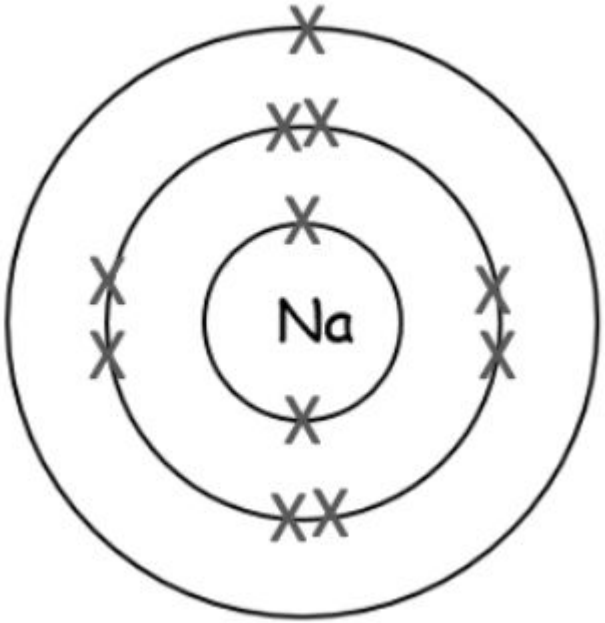


1. What is Sodium electronic configuration?

2. Why is Sodium in Group 1 of the periodic table? (Refer to picture).

3. Why is Sodium in Period 3 of the periodic table? (Refer to picture).

Starter - to discuss



1. What is Sodium electronic configuration? **2,8,1**

2. Why is Sodium in Group 1 of the periodic table? (Refer to picture).

   1 electron on the outermost shell

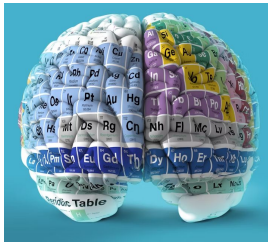
3. Why is Sodium in Period 3 of the periodic table? (Refer to picture).

   It has 3 occupied electron shells

Group 1 elements

Lesson Objectives

- To interpret data to describe patterns in properties of the Group 1 elements.
- To use patterns to predict properties of Group 1 elements.



Brain teaser

What is a group in the periodic table? Tick **1** correct answer

- ☐ A row of elements with the same number of electron shells.
- ✔ A column of elements with the same number of electrons in their outer shell.
- ☐ A row of elements arranged by increasing atomic number.
- ☐ A column of elements with the same number of protons.

Keywords: Periodic Table, group 1, alkali metals, trends, reactive

Group 1 elements		
Meeting	Exceeding	Mastering
I can list the group 1 elements and describe their properties as part of the alkali metals.	I can explain the group 1 elements reactivity in relation to their valence shell electrons	I can explain the trends in the alkali metals as I read up/down the group

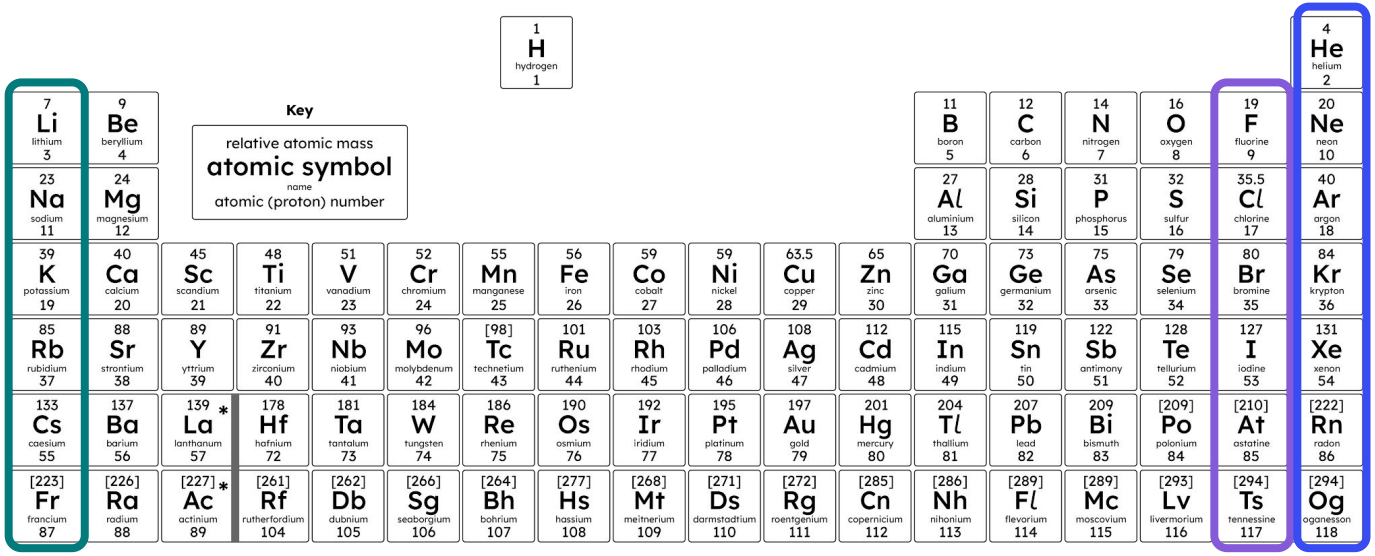
Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Atomic Structure	☐ I can name the subatomic particles and their charges	☐ I can explain how electrons are configured on different atoms	☐ I can use the periodic table to draw the electron configuration for multiple atoms
Group 1 - The Alkali Metals	☐ I can list the group 1 elements and describe their properties as part of the alkali metals.	☐ I can explain the group 1 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the alkali metals as I read up/down the group
Group 7 - The Halogens	☐ I can list the group 7 elements and describe their properties as part of the halogens.	☐ I can explain the group 7 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the halogens as I read up/down the group
Group 0 - The Noble Gases	☐ I can list the group 0 elements and describe their properties as part of the noble gases.	☐ I can explain the group 0 elements are inert due to their valence shell electrons	☐ I can explain the trends in the noble gases as I read up/down the group
Ionic Bonding	☐ I can describe ionic bonding as the combining of a metal atom with a nonmetal atom	☐ I can explain why metals lose electrons to because positive ions and nonmetals gain electrons to become negative ions	☐ I can use the periodic table to draw accurate dot and cross diagrams to demonstrate ionic bonds
Covalent Bonding	☐ I can describe covalent bonding as the combining of two or more nonmetal atoms	☐ I can explain how atoms will share valence electrons and form covalent molecules	☐ I can use the periodic table to draw accurate dot and cross diagrams of single, double or triple covalent bonds

Exploring groups and properties

Explanation

Elements in the same **group** have the same number of electrons on their outer shell. This causes the elements in the same group to have similar chemical properties, showing a trend throughout the group.



																	4 He helium 2
7 Li lithium 3	9 Be beryllium 4											11 B boron 5	12 C carbon 6	14 N nitrogen 7	16 O oxygen 8	19 F fluorine 9	20 Ne neon 10
23 Na sodium 11	24 Mg magnesium 12											27 Al aluminium 13	28 Si silicon 14	31 P phosphorus 15	32 S sulfur 16	35.5 Cl chlorine 17	40 Ar argon 18
39 K potassium 19	40 Ca calcium 20	45 Sc scandium 21	48 Ti titanium 22	51 V vanadium 23	52 Cr chromium 24	55 Mn manganese 25	56 Fe iron 26	59 Co cobalt 27	59 Ni nickel 28	63.5 Cu copper 29	65 Zn zinc 30	70 Ga gallium 31	73 Ge germanium 32	75 As arsenic 33	79 Se selenium 34	80 Br bromine 35	84 Kr krypton 36
85 Rb rubidium 37	88 Sr strontium 38	89 Y yttrium 39	91 Zr zirconium 40	93 Nb niobium 41	96 Mo molybdenum 42	[98] Tc technetium 43	101 Ru ruthenium 44	103 Rh rhodium 45	106 Pd palladium 46	108 Ag silver 47	112 Cd cadmium 48	115 In indium 49	119 Sn tin 50	122 Sb antimony 51	128 Te tellurium 52	127 I iodine 53	131 Xe xenon 54
133 Cs caesium 55	137 Ba barium 56	139 * La lanthanum 57	178 Hf hafnium 72	181 Ta tantalum 73	184 W tungsten 74	186 Re rhenium 75	190 Os osmium 76	192 Ir iridium 77	195 Pt platinum 78	197 Au gold 79	201 Hg mercury 80	204 Tl thallium 81	207 Pb lead 82	209 Bi bismuth 83	[209] Po polonium 84	[210] At astatine 85	[222] Rn radon 86
[223] Fr francium 87	[226] Ra radium 88	[227] * Ac actinium 89	[261] Rf rutherfordium 104	[262] Db dubnium 105	[266] Sg seaborgium 106	[264] Bh bohrium 107	[277] Hs hassium 108	[268] Mt meitnerium 109	[271] Ds darmstadtium 110	[272] Rg roentgenium 111	[285] Cn copernicium 112	[286] Nh nihonium 113	[289] Fl flevorium 114	[289] Mc moscovium 115	[293] Lv livermorium 116	[294] Ts tennessine 117	[294] Og oganesson 118

*Elements with atomic numbers 58-71 and 90-103 have been omitted from this periodic table.

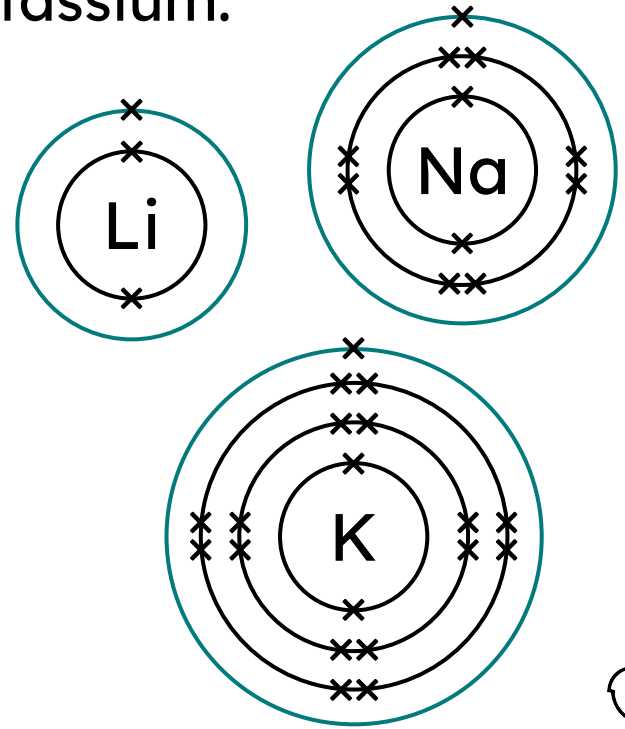
Exploring groups and properties

Explanation

Group 1 is known also as the Alkali metals. This **group** includes lithium, sodium, and potassium.

They have similar chemical and physical properties:

- high reactivity
- react vigorously with cold water
- low melting points



Recall: Complete the table

Practice

Element	Symbol	Atomic No.	Electron Configuration	Valence Electron
Lithium				
Sodium				
Potassium				
Rubidium		37	2,8,8,18,1	
Caesium		55	2,8,8,18,18,1	
Francium		87	2,8,8,18,18,32,1	

Recall: Complete the table

Practice

Element	Symbol	Atomic No.	Electron Configuration	Valence Electron
Lithium				
Sodium				
Potassium				
Rubidium		37	2,8,8,18,1	
Caesium		55	2,8,8,18,18,1	
Francium		87	2,8,8,18,18,32,1	

Recall: Complete the table

Practice

Element	Symbol	Atomic No.	Electron Configuration	Valence Electron
Lithium	Li	3	2,1	1
Sodium	Na	11	2,8,1	1
Potassium	K	19	2,8,8,1	1
Rubidium	Rb	37	2,8,8,18,1	1
Caesium	Cs	55	2,8,8,18,18,1	1
Francium	Fr	87	2,8,8,18,18,32,1	1

How do Group 1 elements react with water?

In this activity you will:

- ★ **record observations** on demonstrations of the reactivity of Group 1 elements with water
- ★ identify a **pattern in behaviour**
- ★ identify the **products formed** and some of the **product properties**.

You will be working scientifically to:

- ★ make observations about what you see
- ★ predict what you would expect to see for other elements in the same group.

Group 1 elements - observations

Practice

Element	Appearance	How easy is it to cut with a knife?	What happens when it is added to water?
lithium			
sodium			
potassium			

Universal indicator solution is added to the water. Why does the indicator change colour?

After the reaction, a lighted splint makes a squeaky pop. What does this show?

Rubidium reacting with water



Caesium reacting with water



Francium reacting with water



Group 1 elements - observations

Practice

Element	Appearance	How easy is it to cut with a knife?	What happens when it is added to water?
lithium			
sodium			
potassium			

Group 1 elements - observations

Practice

Element	Appearance	How easy is it to cut with a knife?	What happens when it is added to water?
lithium	dull and dark grey on outside, shiny on inside	could be cut by knife	forms a sphere and floats on surface, fizzes
sodium			
potassium			

Group 1 elements - observations

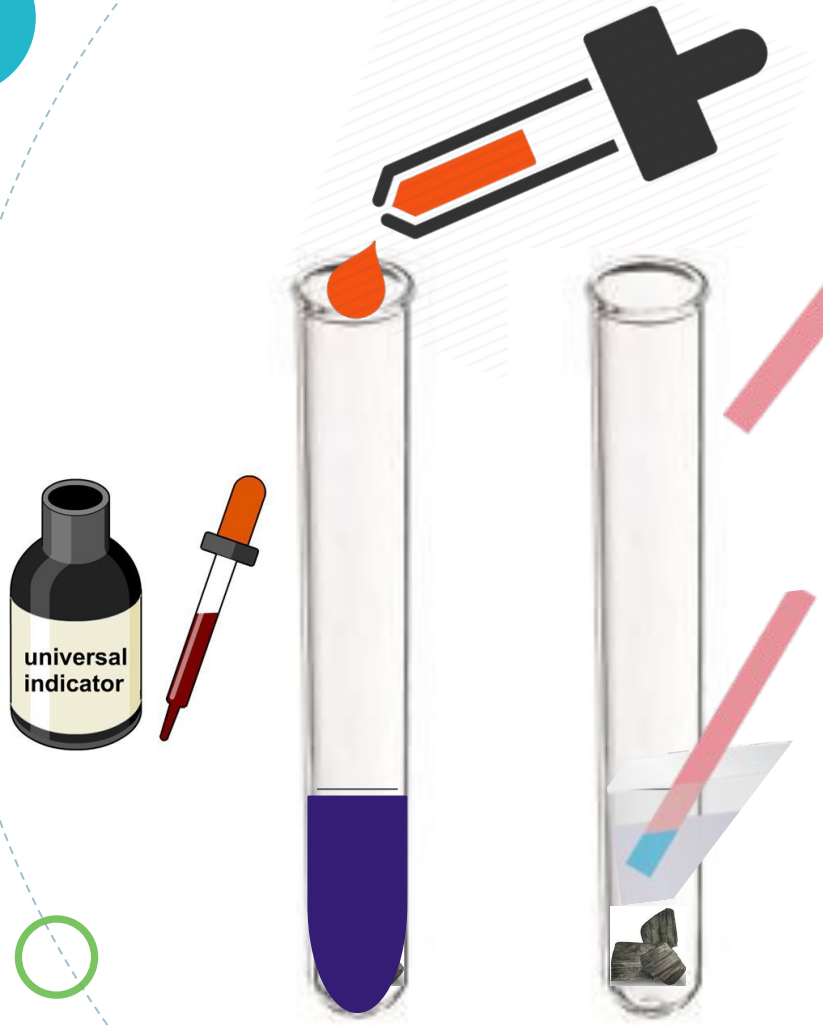
Practice

Element	Appearance	How easy is it to cut with a knife?	What happens when it is added to water?
lithium	dull and dark grey on outside, shiny on inside	could be cut by knife	forms a sphere and floats on surface, fizzes
sodium	dull and light grey on outside, shiny on inside	easier to cut by knife	same as above, but faster and may produce yellow flame
potassium			

Group 1 elements - observations

Practice

Element	Appearance	How easy is it to cut with a knife?	What happens when it is added to water?
lithium	dull and dark grey on outside, shiny on inside	could be cut by knife	forms a sphere and floats on surface, fizzes
sodium	dull and light grey on outside, shiny on inside	easier to cut by knife	same as above, but faster and may produce yellow flame
potassium	dull and grey on outside, shiny on inside	even easier to cut by knife	same as above, even faster, and produce lilac flame (may explode)

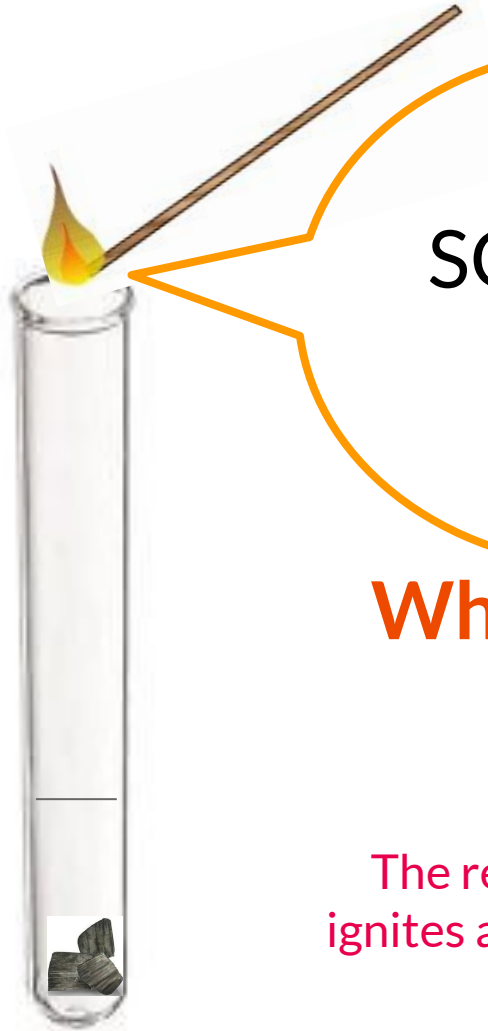


Lithium paper

What does this show?

Alkaline

The reaction produces alkaline hydroxide ions.



SQUEAKY POP!

What does this show?

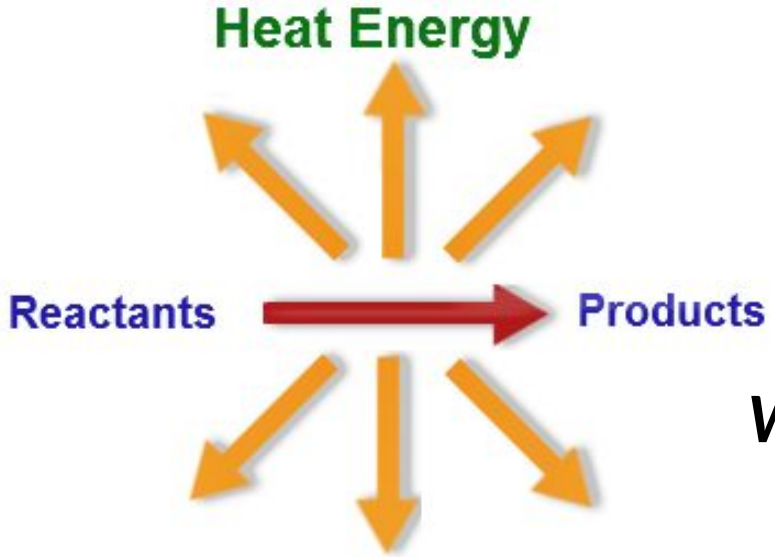
Hydrogen

The reaction produces hydrogen that ignites a lighted splint with a squeaky pop

Group 1 metal + water → metal hydroxide + hydrogen

lithium + water → lithium hydroxide + hydrogen

Exothermic reaction

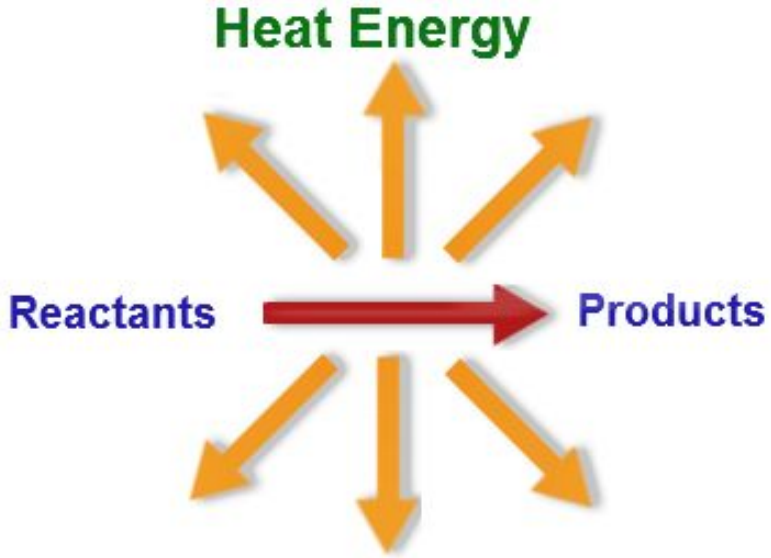


The reactions **produce heat energy**.

When potassium reacts with water, it produces so much heat that the hydrogen given off catches alight.

What colour was the flame?

Exothermic reaction



The reactions **produce heat energy**.

When potassium reacts with water, it produces so much heat that the hydrogen given off catches alight.

What colour was the flame?

Lilac flame

Group 1: Alkali metals - Discuss



ALL

Describe the trend in the reactivity of the elements down Group 1.

MOST

Write the word equation for the reaction between lithium and water.

SOME

Suggest a reason why the alkali metals are stored in oil.

Group 1: Alkali metals

Practice

ALL

The metals become more reactive as you go down the group.

MOST

lithium + water → lithium hydroxide + hydrogen



SOME

The metals are highly reactive with water and oxygen in the air, but they do not react with oil. This prevents them from reacting to form compounds, allowing them to remain as pure metals.

Element	Chemical formula	Atomic number	Melting point (℃)	Boiling point (℃)
Lithium	Li	3	180	1360
Sodium	Na	11	98	900
Potassium	K	19	63	777

What is the trend shown in the boiling points of Group 1 elements?

What prediction can you make about the boiling point of rubidium?

How many more electrons than lithium does an atom of sodium have?

Element	Chemical formula	Atomic number	Melting point (℃)	Boiling point (℃)
Lithium	Li	3	180	1360
Sodium	Na	11	98	900
Potassium	K	19	63	777

What is the trend shown in the boiling points of Group 1 elements?

The boiling points decrease, as you go down the group.

What prediction can you make about the boiling point of rubidium?

The boiling point of rubidium will be less than 777℃.

How many more electrons than lithium does an atom of sodium have?

8 more electrons.

acid	calcium	down	hard	hydrogen
less	lithium	magnesium	more	not
up	potassium	reactive	soft	unreactive
sodium	oxygen	easily	oil	

The first three elements in Group 1 are ....., ... and ..... .

The first three elements in Group 1 are **lithium**, **sodium** and **potassium** .

acid	calcium	down	hard	hydrogen
less	lithium	magnesium	more	not
up	potassium	reactive	soft	unreactive
sodium	oxygen	easily	oil	

The Group 1 elements are very ..... .
They can ..... be cut with a knife.

The Group 1 elements are very **soft** . They can **easily** be cut with a knife.

acid	calcium	down	hard	hydrogen
less	lithium	magnesium	more	not
up	potassium	reactive	soft	unreactive
sodium	oxygen	easily	oil	

The Group 1 elements are stored under ..... because they are very ..... .

The Group 1 elements are stored under **oil** because they are very **reactive** .

acid	calcium	down	hard	hydrogen
less	lithium	magnesium	more	not
up	potassium	reactive	soft	unreactive
sodium	oxygen	easily	oil	

When these elements react with water they form the gas ..... .

When these elements react with water they form the gas **hydrogen** .

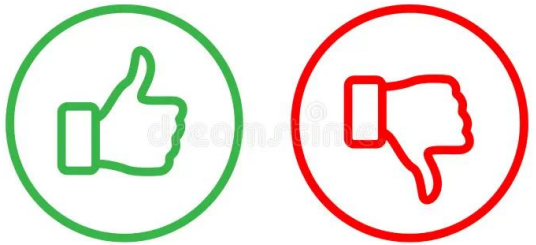
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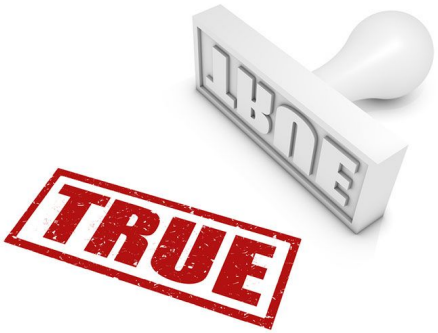
The elements are ..... reactive with water as you go ..... the group.

The elements are **more/less** reactive with water as you go **down/up** the group.

TRUE OR FALSE

Thumbs up or down to show '**true**' or '**false**' for the following statements.





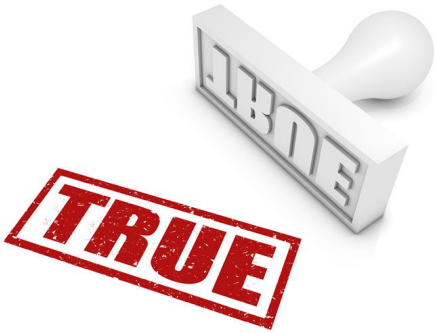
Group 1 elements are metals.



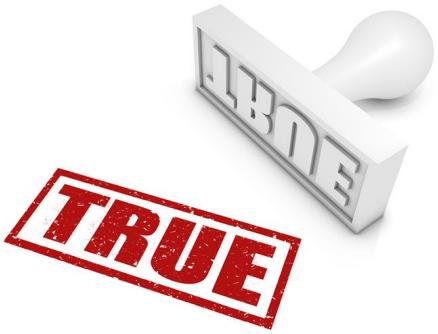
Group 1 elements are hard.



Group 1 elements become less reactive down the group.



Group 1 elements can conduct heat and electricity.



Group 1 elements react with water to produce metal hydroxides and hydrogen.



- Group 1 elements' boiling points increase down the group.

Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Atomic Structure	☐ I can name the subatomic particles and their charges	☐ I can explain how electrons are configured on different atoms	☐ I can use the periodic table to draw the electron configuration for multiple atoms
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Group 7 - The Halogens	☐ I can list the group 7 elements and describe their properties as part of the halogens.	☐ I can explain the group 7 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the halogens as I read up/down the group
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